

Estimating the Degree of Mental State using Heart Rate while Studying

Yusuke Kawasaki
Aoyama Gakuin University
Sagamihara, Japan

Anna Yokokubo
Aoyama Gakuin University
Sagamihara, Japan
yokokubo@it.aoyama.ac.jp

Tahera Hossain
Aoyama Gakuin University
Sagamihara, Japan
tahera@wil-aoyama.jp

Guillaume Lopez
Aoyama Gakuin University
Sagamihara, Japan
guillaume@it.aoyama.ac.jp

ABSTRACT

The purpose of this study is to develop a system that continuously estimates the mental state during study time using a smartwatch equipped with a heart rate sensor and supports improvement in performance through short-term actions and long-term feedback. We verified whether different mental states (e.g., degree of focus, drowsiness, and fatigue) could be sufficiently detected during the study time using heart rate-related features from the heart rate sensor of the smartwatch. Firstly, we compared the performance of evaluation having only heart rate data for a different time window. Afterwards robustness evaluation has done for the unknown users' data. Extra Trees Classifier fed with features extracted from 10 minutes window got the highest estimation accuracy of focus, drowsiness, and fatigue. Each mental state degree could be estimated with an average F1-Score value of more than 78.8%, which states that heart rate features may be sufficient to assess a certain degree of mental state.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing.

KEYWORDS

Mental State; Heart rate; Machine Learning

ACM Reference Format:

Yusuke Kawasaki, Tahera Hossain, Anna Yokokubo, and Guillaume Lopez. 2021. Estimating the Degree of Mental State using Heart Rate while Studying. In *Adjunct Proceedings of the 2021 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2021 ACM International Symposium on Wearable Computers (UbiComp-ISWC '21 Adjunct)*, September 21–26, 2021, Virtual, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3460418.3479317>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

UbiComp-ISWC '21 Adjunct, September 21–26, 2021, Virtual, USA

© 2021 Association for Computing Machinery.

ACM ISBN 978-1-4503-8461-2/21/09...\$15.00

<https://doi.org/10.1145/3460418.3479317>

1 INTRODUCTION

There are various situations where focus is required in daily life, such as studying, working, or playing sports. In particular, it is one of the most important factors when studying, as it is something that anyone must do at any age. High focus is required to enhance the effectiveness of studying. However, despite the importance of high focus, quantitatively evaluating the high focus is not widespread in daily life. In addition, in modern times, study time is often considered more important than study quality. According to the results of the 18th 21st Century Longitudinal Survey of Births by the Ministry of Education, Culture, Sports, Science and Technology, students who are in the third year of high school who are going to college or graduate school often spend more than 6 hours studying on holidays (26.7% for university applicants and 49.5% for graduate school applicants)[7]. Also, in the questionnaire, there are only questions about study time. From this, it is thought that study time is one of the evaluations in modern times. However, it has been shown that study efficiency is improved by cumulative learning with short breaks at regular intervals compared to long-term learning with no breaks.[12]. According to a Benesse Corporation study, a 15-minute $\times 3$ (total 45 minutes) stacked learning compared to 60-minute long learning indicates that stacked learning has a better score than long-term learning, even though the study time is less [3]. In addition, the gamma wave of the frontal lobe, which is involved in focus, decreased sharply after 40 minutes, suggesting that focus may last only about 40 minutes.

This research aims to continuously estimate the mental state during study using wearable devices and realize a system that supports performance improvement through short-term action and long-term feedback. We quantify the degree of numerous mental states during the study time by utilizing simply the heart rate data of a smartwatch, which is easy to reproduce in daily life. We have collected heart rate data (HR) and different mental state data during study time for this experiment. We investigated whether the heart rate alone could achieve sufficient accuracy to detect mental state during study time. We explore the Support Vector Machine (SVM), K-Neighbor Method (KNN), Random Forest (RF), Gradient Boosted Regression Trees (GBRT), and Extra Trees Classifier (ET) for classification. Each mental state degree could be estimated with an average F1-Score value of more than 78.8%, which states that heart rate features may be sufficient to assess a certain degree of mental state.

2 PREVIOUS RESEARCH ON MENTAL STATE ESTIMATION

Okubo et al.[8] proposed a system to evaluate a concentrated state by attaching an acceleration sensor to a chair. In the experiment, three subjects were asked to type for 30 minutes. The relationship between the sum of the power spectrum of the acceleration and the number of keystrokes per unit time was examined. The results show a strong negative correlation between the sum of power spectra and the number of keystrokes per unit time. As the sum of the power spectra becomes smaller, the estimation of the focus becomes valid. However, the chair used in the experiment is only the rotating chair, and it has been tested only in a limited environment. And, there was a problem that it was difficult only by the user himself to correctly attach the sensor to the chair which the user utilizes.

Various studies of psychology and neuroscience have shown that Electroencephalogram (EEG) can be used as indicators of mental state [10]. To estimate the human mental state, the ratio of α wave and β wave obtained by discrete Fourier transform on the EEG signal and it is effective to measure the power spectrum of α wave and β wave [5][2]. Generally, the α wave is the waveform observed in resting and waking, and the amplitude of the α wave increases in the relaxed state. On the other hand, the α wave amplitude decreases, and the β wave appears during tension. Hirai et al.[4] comparatively analyzed the power spectrum of α wave, β wave, low γ wave, and β/α by the experiment for correlation analysis of thought and memory in the learning condition using the simple electroencephalograph. Observing the beta/alpha component, it was shown to be an index to measure the degree of stress and the degree of focus for momentary thinking. It was also stated that low γ is an adequate criterion for measuring the degree of memory. However, there are problems such as that it is difficult to utilize in the daily life, because of the size of the device, difficulty of mounting the device only by the person himself, and sensitivity to the effect of the electromagnetic noise.

Estimation of mental fatigue state using electrocardiographic (ECG) sensors is also widely studied nowadays. Heart rate variability refers to fluctuations in the RRI (R-R Interval), which is the interval between narrow upwards waves (R-waves) on cardiac potentials measured during the heartbeat. These results suggest that the sympathetic nervous system of the autonomic nervous system is activated when the fluctuation of RRI is small and the parasympathetic nervous system is activated when the fluctuation of RRI is large [1].

In the research of Tsunoda et al.[11], Kana et al.[9], it has been confirmed that there is a correlation between the feature quantity of the heart rate variability (HRV) and the mental fatigue level. It has been indicated in many types of research that sequential estimation in mental fatigue is possible. However, it is necessary to continuously measure heart rate intervals with high accuracy to measure heart rate fluctuations, and it is still impossible to use high-grade devices in daily life.

Smartwatches do not exactly measure HR but pulse rate (PR) from the Pulse-to-Pulse Interval (PPI), which is the interval between pulse wave peaks when peripheral blood flow is measured. Although features of the periodic fluctuation of PPI (Pulse-to-Pulse Interval) are not exactly the same as those of heart rate interval

(RRI), they also represent mental states such as emotion and emotion, the periodic fluctuation of PPI (PRV)[13]. In this study we use smartwatch's PR value as an equivalent of HR since it can be measured from the wrist in a less invasive manner.

3 ESTIMATION OF MENTAL STATE DEGREE USING HEART RATE DATA

3.1 Experimental settings

As shown in Fig. 1, subjects worn a smartwatch equipped with a photoelectric plethysmograph sensor (FitBit Versa 2) on either arm to measure heart rate while carrying on actual homework. Each degree of focus, drowsiness, and fatigue was set to three levels of low, medium, and high, and the subjects manually recorded in the application of Fig. 2 when the degree of either mental status label changed at the start of study and during study. Heart rate data during study and degree of mental status label data were obtained according to the following procedure of Fig. 3.



Figure 1: Smartwatch attached on the left arm

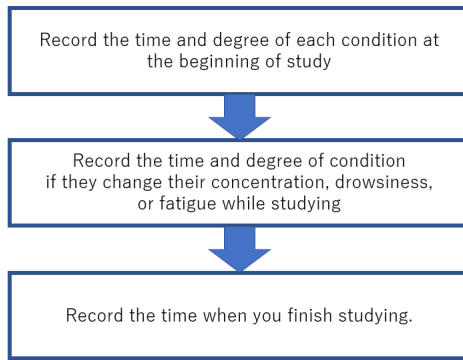


Figure 2: Screenshots of Smartwatch application used to record HR and mental state changes: start HR recording screen (left) and label recording screen (right)

Five male and female high school students attending cram schools cooperated to obtain continuous heart rate data and degree data of mental status labels during study at home. Subjects followed the same procedure as in the Figure 3 and studied as usual.

3.2 Heart Rate Data Collection

We have collected heart rate data and different mental state data from 5 subjects while they were studying. Experiments were conducted from July 31, 2021 to August 30, 2021. Since the characteristics of HR variations due to mental state are not clear, it is

**Figure 3: Experimental procedure**

necessary to investigate the optimal time interval reflecting the relation between HR variations and changes of mental state situation. To determine the optimal time interval, we separated data in the following three ways:

- 3 minutes window shift 20 seconds data set
- 5 minutes window shift 30 seconds data set
- 10 minutes window shift 60 seconds data set

Table 1. is the explanation of the degree of mental state. Therefore, the mental state situations in different interval data set are compared for result analysis. The heart rate feature set and the number of label data collected for each mental status were summarized in Table 2. In our dataset, the highest number of samples is 18,707 with 3 minutes time window, and the lowest number of samples is 5525 with the 10 minutes time window.

Table 1: Explanation of the degree of mental state.

Mental state	Degree	Explanation
CON	LOW	I haven't studied at all.
	MID	I don't think I can concentrate.
	HIGH	Usually. I may not have been able to concentrate a little.
DRO	LOW	I am studying very well.
	MID	I feel like I was very focused.
	HIGH	I'm not sleepy at all.
FAT	LOW	I'm a little sleepy,
	MID	but it doesn't hinder my studies.
	HIGH	Very sleepy. I'm about to fall asleep.
FAT	LOW	I can afford it at all.
	MID	I think I can continue,
	HIGH	but I feel tired from studying.
FAT	LOW	I want to stop right now.
	MID	I can't think of it anymore.
	HIGH	

3.3 Analysis methods and results

Two analytical methods were performed: evaluation of window time and learning model is described in 3.3.1, and robustness evaluation for unknown users is described in 3.3.2.

Table 2: Collected Heart rate and label data. Here FOC, DRO and FAT represents degree of focus, drowsiness, and fatigue respectively.

Window (minutes)	Mental state	Degree of mental state		
		LOW	MID	HIGH
3	FOC	1240	11568	5899
	DRO	12766	4626	1315
	FAT	9484	7966	1257
5	FOC	800	7455	3816
	DRO	8221	2998	852
	FAT	6109	5149	813
10	FOC	366	3399	1760
	DRO	3737	1387	401
	FAT	2776	2376	373

3.3.1 Window Time and Learning Model Evaluation. Seven features were extracted from each time window heart rate data for analysis: mean, maximum, minimum, maximum-minimum, standard deviation, kurtosis, and skewness. 75% randomly selected from the dataset was used for learning. Five models, Support Vector Machine (the hyper-parameters are $C = 7.0$ and $\gamma = 3.0$) (SVM), K-Neighbor Method, (the hyper-parameter is $n_neighbors = 1$) (KNN), Random Forest (the hyper-parameters are $max_depth = None$, $max_leaf_nodes = None$, $n_estimators = 175$ and $random_state=0$) (RF), Gradient Boosted Regression Trees (the hyper-parameter is $random_state=0$) (GBRT), and Extra Trees Classifier (the hyper-parameters are $n_estimators=100$, $random_state=0$) (ET), were learned and evaluated with the remaining 25% data. All classifiers were analyzed with initial parameters. The process was repeated ten times, and the F1-Score values (%) and standard deviations of each mental state were calculated.

Table 5, Table 4, Table 3 are the results of the analysis. The highest F1-Score of focus, drowsiness, and fatigue were found in the 10-minute window data set and the ET model, and the mental state of each of them could be estimated with an average F1-Score value of more than 78.8% (Table 6). Besides, regarding the effect of HR data window size on mental state estimation accuracy, the larger the window is, the higher the F1-score, whichever the classifier. Having a closer look at the results, we can observe that for the mental states from which a large number of data have been collected, the accuracy using a random forest model is close to the one obtained with extra-trees classifier (e.g. moderate focus, low drowsiness).

3.3.2 Robustness Evaluation for Unknown Users. Next, the F1-Score evaluation for the unknown user data was carried out using the above-described heart rate feature quantity and learning model of the classifier. Among the data of the five persons mentioned above, after learning the model by removing the data of one person in order, the test was carried out with the data of removed user.

Table 7 and Table 8 show the results of the analysis. Obviously, results are less accurate than for the first analytical method. Labels that exceeded 60% fell below 10%, and labels that exceeded 80% were also significantly less accurate. There is no apparent difference between the 10-minute window, the 5-minute window, and the 3-minute window.

Table 3: Learning Model Evaluation Results (10min Window)

	LOW	MID	HIGH	AVERAGE
FOC(ET)	74.2±3	88.8±0	79.4±1	80.8
DRO(ET)	91.2±1	77.7±1	67.5±5	78.8
FAT(ET)	87.7±1	84.3±1	73.7±3	81.9
FOC(RF)	66.7±5	86.6±1	74.0±2	75.8
DRO(RF)	89.2±1	70.4±2	59.7±6	73.1
FAT(RF)	84.9±1	81.7±1	64.8±3	77.1
FOC(KNN)	64.0±3	82.9±1	70.4±2	72.4
DRO(KNN)	86.0±1	67.7±2	54.0±3	69.2
FAT(KNN)	81.1±1	77.3±1	60.6±5	73.0
FOC(SVM)	55.2±4	82.5±1	68.1±2	68.6
DRO(SVM)	85.9±1	63.6±2	50.4±5	66.6
FAT(SVM)	79.8±1	76.4±1	54.9±5	70.4
FOC(GBRT)	27.2±4	79.2±1	47.9±4	51.4
DRO(GBRT)	83.0±1	34.7±2	26.6±3	48.1
FAT(GBRT)	73.5±2	68.2±1	24.1±6	55.3

Table 4: Learning Model Evaluation Results (5min Window)

	LOW	MID	HIGH	AVERAGE
FOC(ET)	61.2±3	85.3±1	71.6±1	72.7
DRO(ET)	87.8±0	67.5±1	59.2±3	71.5
FAT(ET)	81.8±1	77.4±0	63.7±3	74.3
FOC(RF)	53.7±2	83.5±0	65.9±1	67.7
DRO(RF)	86.3±0	60.7±1	49.1±3	65.4
FAT(RF)	79.6±1	75.7±1	55.8±2	70.4
FOC(KNN)	50.5±1	78.6±0	62.2±1	63.8
DRO(KNN)	81.7±1	57.2±1	46.3±2	61.7
FAT(KNN)	75.5±1	70.3±1	51.3±3	65.7
FOC(SVM)	40.8±2	79.2±1	58.4±1	59.5
DRO(SVM)	82.9±0	52.0±1	33.5±2	56.1
FAT(SVM)	74.6±1	70.9±1	40.7±3	62.1
FOC(GBRT)	17.0±3	77.5±0	31.1±2	41.9
DRO(GBRT)	81.7±0	18.6±1	13.8±3	38.0
FAT(GBRT)	69.5±1	63.2±1	20.3±5	51.0

4 DISCUSSION

Although there were some items under 70% in the 10-minute window data set and ET model in which the F1-Score was the best for the model learned by the data set not depending on the individual, the low focus of about 400 records, high fatigue, and high drowsiness became the lowest F1-Score respectively, when it was compared with the data number of each condition degree, and the medium focus, low fatigue and low drowsiness with many records became the high accuracy which respectively exceeded 87%. From this fact, it seems to be able to expect an improvement in the F1 value, if further data is collected, and if the imbalance of every label is reduced. From this fact, it was indicated that the degree of mental state could be estimated to some extent only by the heart rate.

In addition, while the analysis results with models trained on non-individual datasets showed significant differences with time windows and classifiers, the results of the analysis of unknown

Table 5: Learning Model Evaluation Results (3min Window)

	LOW	MID	HIGH	AVERAGE
FOC(ET)	50.7±2	80.6±0	60.8±1	64.0
DRO(ET)	84.3±0	54.8±1	45.3±2	61.5
FAT(ET)	77.2±0	71.1±0	50.0±2	66.1
FOC(RF)	40.1±2	79.5±0	55.3±1	58.3
DRO(RF)	83.6±0	49.2±1	36.9±2	56.6
FAT(RF)	75.5±1	70.5±1	43.6±2	63.2
FOC(KNN)	36.9±2	73.5±0	51.7±1	54.0
DRO(KNN)	77.8±0	47.6±1	34.0±2	53.1
FAT(KNN)	69.6±1	63.8±1	38.3±1	57.2
FOC(SVM)	25.5±3	76.0±0	47.0±1	49.5
DRO(SVM)	80.5±1	37.7±1	25.1±3	47.8
FAT(SVM)	69.6±1	65.0±1	27.6±2	54.1
FOC(GBRT)	11.7±1	76.7±0	21.3±2	36.6
DRO(GBRT)	81.2±0	10.7±1	9.4±2	33.8
FAT(GBRT)	66.4±1	61.6±1	13.6±2	47.2

Table 6: Average mental state estimation accuracy of the best learning model for each window size

Window size (min)	3	5	10
Model	ET	ET	ET
FOC	64.0	72.7	80.8
DRO	61.5	71.5	78.8
FAT	66.1	74.3	81.9

Table 7: Results of robustness evaluation for unknown users (10min Window)

	LOW	MID	HIGH
FOC(ET)	1.0	62.2	32.4
DRO(ET)	68.8	15.0	2.5
FAT(ET)	39.9	51.5	1.7
FOC(RF)	1.8	60.8	37.2
DRO(RF)	65.4	16.2	3.7
FAT(RF)	36.6	53.8	2.0
FOC(KNN)	9.6	53.6	38.7
DRO(KNN)	64.5	27.2	2.5
FAT(KNN)	33.7	52.1	7.4
FOC(SVM)	5.3	57.8	34.8
DRO(SVM)	66.2	18.2	1.9
FAT(SVM)	42.9	48.8	1.2
FOC(GBRT)	7.1	54.2	39.4
DRO(GBRT)	57.5	26.5	5.5
FAT(GBRT)	29.3	57.5	5.9

data with models trained on non-individual datasets showed no significant differences other than GBRT from 40% to 50%. Since the learning of each model was insufficient, it seems to have become low accuracy in all models. This fact shows that the estimation for the unknown data did not work. However, we can expect improvements by calibration for each individual.

Table 8: Results of robustness evaluation for unknown users (3min Window)

	LOW	MID	HIGH
FOC(ET)	2.8	61.1	30.1
DRO(ET)	66.0	24.2	0.1
FAT(ET)	41.2	54.4	7.0
FOC(RF)	6.6	55.9	33.2
DRO(RF)	60.1	30.2	2.6
FAT(RF)	37.8	56.6	7.0
FOC(KNN)	9.0	54.8	34.9
DRO(KNN)	57.5	32.0	3.4
FAT(KNN)	37.4	53.1	9.3
FOC(SVM)	10.1	54.9	34.5
DRO(SVM)	58.8	32.5	2.2
FAT(SVM)	37.5	55.1	8.7
FOC(GBRT)	12.4	48.2	39.2
DRO(GBRT)	36.5	39.3	8.7
FAT(GBRT)	19.8	59.7	10.3

5 CONCLUSION AND FUTURE WORK

In this work, we attempted to estimate the degree of multiple mental states during study at home using only the heart rate data of a smartwatch, which is easy to reproduce in daily life. Using the heart rate data, two analysis methods were performed. First, we evaluated the window time and the learning model. The most accurate time window was ten minutes, whatever the learning model used. On the other hand, the optimal learning model was an extra-trees classifier, whatever the time windows used, which best average F1-Score reached 78.8% with 10 minutes window time. Second, we performed a robustness evaluation to evaluate the accuracy of data collected from unknown users. The results were lower than obtained by the first analysis method. However, we believe that improvements can be expected by individual calibration, as shown in previous studies[6]. As a prospect, we will improve by collecting more data, optimizing the learning model, and implementing feature quantity engineering.

ACKNOWLEDGMENTS

This study is the result of a joint research between JSOL Co., Ltd. and the authors. JSOL Co., Ltd., has collected the experimental data of this research at a partner cram school.

REFERENCES

- [1] U Rajendra Acharya, Paul Joseph, N Kannathal, Choo Lim, and Jasjit Suri. 2007. Heart rate variability: A review. *Medical and biological engineering and computing* 44 (01 2007), 1031–51. <https://doi.org/10.1007/s11517-006-0119-0>
- [2] V. Amaral, Leonardo Ferreira, Plinio Aquino Junior, and Maria Claudia Castro. 2013. EEG signal classification in usability experiments. *ISSNIP Biosignals and Biorobotics Conference, BRC*, 1–5. <https://doi.org/10.1109/BRC.2013.6487469>
- [3] Benesse. 2017. Learning time divided into 45 minutes demonstrate same or greater learning effect than 60 minutes - Short-time intensive "stacked learning" was more effective than "long-term learning". Retrieved January 09, 2021 [in Japanese] from <https://prtimes.jp/main/html/rd/p/000000562.000000120.html>
- [4] F. Hirai, Kouji Yoshida, and I. Miyaji. 2013. Comparison analysis of the thought and the memory at the learning time by the simple electroencephalograph. *Multimedia, Distributed, Cooperative, and Mobile Symposium (DICO) 2013, Symposium Proceedings* (01 2013), 1441–1446.
- [5] Hitoshi Masaki, Masao Ohira, Hidetake Uwano, and Ken-ichi Matsumoto. 2011. A Quantitative Evaluation on the Software Use Experience with Electroencephalogram. 469–477. https://doi.org/10.1007/978-3-642-21708-1_53
- [6] Kizito Nkurikiyeyezu, Anna Yokokubo, and Guillaume Lopez. 2020. The Influence of Person-specific Biometrics in Improving Generic Stress Predictive Models. *Sensors and Materials* 32 (2020), 703–722.
- [7] Japan Ministry of Education Culture Sports Science and Technology. 2020. 18th 21st Century Longitudinal Survey of Births. Retrieved May 04, 2021 [in Japanese] from https://www.mext.go.jp/content/20200902-mxt_chousa01-000009574_1rrrrr.pdf
- [8] Masashi Okubo and Aya Fujimura. 2009. A proposal of estimation system for concentration situation by using acceleration sensor. In *RO-MAN 2009 - The 18th IEEE International Symposium on Robot and Human Interactive Communication*. 498–503. <https://doi.org/10.1109/ROMAN.2009.5326219>
- [9] Kana Shoji, Shun Ishii, Anna Yokokubo, and Guillaume Lopez. 2019. Mental fatigue estimation in desk work using biological signals. *DICOMO* (2019).
- [10] Shun Tanabe, Ryosuke Nagase, Kosuke Fujita, Hideaki Touyama, and Yoshito Tobe. 2017. HOKAGE: Providing the attention level of a remote user using EEG. In *2nd International Workshop on Smart Sensing Systems (IWSSS)*.
- [11] Keisuke Tsunoda, Akihiro Chiba, Hiroshi Chigira, Tetsuya Ura, and Osamu Mizuno. 2015. Estimating changes in a cognitive performance using heart rate variability. In *2015 IEEE 15th International Conference on Bioinformatics and Bioengineering (BIBE)*. 1–6. <https://doi.org/10.1109/BIBE.2015.7367712>
- [12] Yusuke Watanabe and Yuji Ikegaya. 2017. Effect of intermittent learning on task performance: a pilot study. *Journal of Neurosci* 38 (2017), 1–5.
- [13] Emi Yuda, Muneichi Shibata, Yuki Ogata, Norihiro Ueda, Tomoyuki Yambe, Makoto Yoshizawa, and Junichiro Hayano. 2020. Pulse rate variability: a new biomarker, not a surrogate for heart rate variability. *J Physiol Anthropol* 39, 21 (2020).